

Problem 2.3

Consider the amount function

$$A(t) = t^2 + 2t + 3.$$

1. Find the corresponding accumulation function.
2. Find I_n in terms of n .

Solution

(a) Accumulation function

The accumulation function is defined by

$$a(t) = \frac{A(t)}{A(0)}.$$

Since

$$A(0) = 0^2 + 2(0) + 3 = 3,$$

we have

$$a(t) = \frac{t^2 + 2t + 3}{3}.$$

Therefore,

$$\boxed{a(t) = \frac{t^2}{3} + \frac{2t}{3} + 1}.$$

(b) Find I_n

The interest earned in the n th period is

$$I_n = A(n) - A(n-1).$$

Now,

$$A(n) = n^2 + 2n + 3,$$

and

$$A(n-1) = (n-1)^2 + 2(n-1) + 3.$$

Expand $A(n-1)$:

$$A(n-1) = n^2 - 2n + 1 + 2n - 2 + 3 = n^2 + 2.$$

Thus,

$$I_n = (n^2 + 2n + 3) - (n^2 + 2).$$

So,

$$I_n = 2n + 1.$$

Therefore,

$$\boxed{I_n = 2n + 1}.$$